



5CS037 - Concepts and Technologies of AI

REGRESSION TASK

Predicting Global Electricity Demand (TWh)

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GitHub link- https://github.com/Shubhangad-Shrestha/Final_Portfolio_AI

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Abstract

Purpose: This report predicts national and regional electricity demand (TWh) from macro-level population, economic, and energy-consumption indicators using regression techniques, supporting SDG 7 - Affordable and Clean Energy.

Dataset: The dataset is derived from the Our World in Data (OWID) Global Energy dataset, originally containing 23,377 records and 130 features. Twelve relevant features plus the target variable were selected, and after removing records with a missing target and imputing missing feature values with the median, the final cleaned dataset contained 6,378 records and 12 features (11 predictors and 1 target).

Approach: The methodology included exploratory data analysis (EDA) with distribution and correlation analysis, three regression models (a Multi-Layer Perceptron Neural Network, Ridge Regression, and Random Forest Regressor), hyperparameter optimization using GridSearchCV with 5-fold cross-validation, and SelectKBest (f_regression) feature selection.

Key Results: The Random Forest Regressor achieved the best overall performance (Test $R^2 = 0.640$, Test RMSE = 1,597.83 TWh), followed by the Neural Network (Test $R^2 = 0.491$) and Ridge Regression (Test $R^2 = 0.488$). Primary energy consumption, GDP, and oil consumption showed the strongest correlation with electricity demand.

Conclusion: Unlike the earlier iteration of this project, which used a very small (63-record) dataset and produced negative R^2 scores for every model, the larger OWID-derived dataset (6,378 records) allowed all three models to explain a meaningful proportion of the variance in electricity demand. The Random Forest model is recommended as the final model due to its highest test R^2 and lowest test error, although some overfitting (Train $R^2 = 0.790$ vs Test $R^2 = 0.640$) remains a consideration for future refinement.

1. Introduction

1.1 Problem Statement

The goal of this project is to predict electricity demand (measured in terawatt-hours, TWh) using macro-level indicators such as population, GDP, and the consumption of different primary energy sources (coal, oil, gas, renewables, solar, wind, and hydro). Accurate prediction of electricity demand supports better grid planning, energy policy formulation, infrastructure investment decisions, and progress tracking toward global clean-energy targets. This directly supports SDG 7: ensuring access to affordable, reliable, sustainable, and modern energy for all.

1.2 Dataset

The dataset used in this analysis is the Our World in Data (OWID) Global Energy dataset, a publicly available and continuously updated dataset compiled by the OWID team from sources including the Energy Institute Statistical Review of World Energy, Ember, and the United Nations. It was accessed directly from the OWID GitHub repository ([owid/energy-data](https://github.com/owid/energy-data)) as a CSV file containing 23,377 rows and 130 columns spanning individual countries, regions, and world aggregates from 1900 to 2025.

For this regression task, 13 columns were selected: country, year, population, gdp, primary_energy_consumption, renewables_consumption, coal_consumption, oil_consumption, gas_consumption, solar_consumption, wind_consumption, hydro_consumption, and the target variable electricity_demand (TWh). This dataset aligns with SDG 7 because it links population, economic activity, and the energy mix directly to demand for electricity, supporting evidence-based clean-energy planning and monitoring of the global energy transition.

Questions this dataset can help answer include:

- How strongly do population and GDP growth drive electricity demand across countries and time?
- Can the share of consumption from renewable sources (solar, wind, hydro) explain variation in electricity demand independently of fossil-fuel consumption?
- Which combination of energy-mix and macroeconomic features best predicts short-term electricity demand for planning purposes?

Dataset quality assessment revealed significant missingness: population (19.2%), GDP (49.6%), primary energy consumption (43.3%), and the remaining consumption columns and the target variable (72.6% each) contained missing values, largely because many rows correspond to years, countries, or aggregates for which certain data was never collected. No duplicate rows were found.

1.3 Objective

The objective of this analysis is to build and compare regression models (a Neural Network, Ridge Regression, and Random Forest Regressor) that estimate electricity demand from population, economic, and energy-mix features; to identify the most influential predictors through systematic feature selection; to optimize model performance via hyperparameter tuning with cross-validation; and to critically compare the resulting models on unseen test data.

2. Methodology

2.1 Data Preprocessing

The dataset was cleaned in the following order: (1) duplicate rows were checked and removed (none were found); (2) the 16,999 rows (72.7%) with a missing target value (electricity_demand) were dropped, since imputing the target itself would bias model evaluation; (3) the remaining missing values in the numeric predictor columns were imputed using the column median, for example population (322 values, median $\approx$ 7.39 million), GDP (2,248 values, median $\approx$ \$82.9 billion), and the six energy-consumption columns (3,810 values each). After cleaning, the dataset contained 6,378 records with zero remaining missing values and zero duplicates. The non-numeric country column was retained for context but excluded from model inputs, since a categorical country code would introduce high-cardinality noise inappropriate for this stage of the pipeline. Features were split into an 80/20 train-test split (5,102 training rows, 1,276 test rows), and StandardScaler was applied to the features (and, for the Neural Network, to the target as well) prior to training the Neural Network and Ridge Regression models; the Random Forest model was trained on the unscaled features, since tree-based models do not require feature scaling.

2.2 Exploratory Data Analysis (EDA)

Exploratory analysis examined the distribution of the target variable and its relationship with the predictor features.

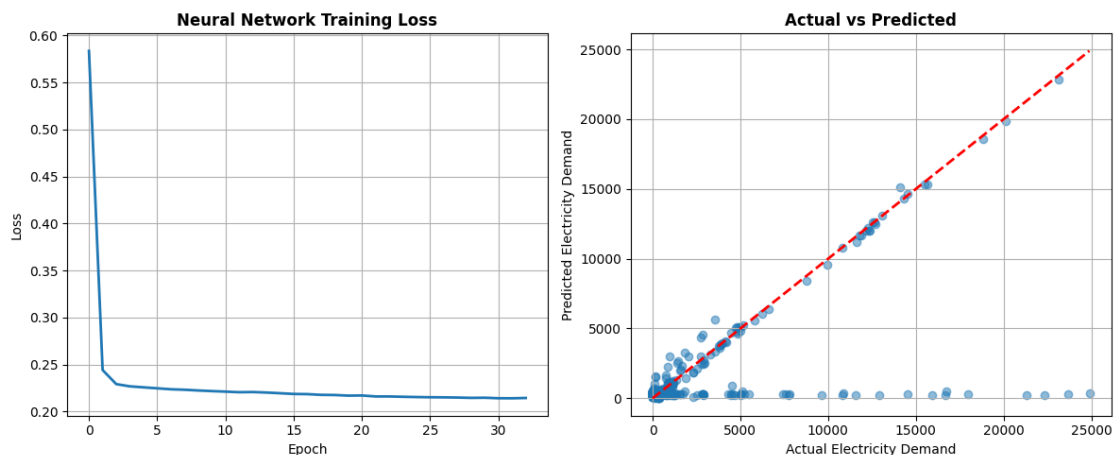


Figure 1: Distribution and box plot of electricity demand (TWh), showing a strongly right-skewed distribution with a small number of very large outliers (large countries and world/regional aggregates).

Electricity demand ranges from 0 to 31,772.35 TWh, with a mean of 647.69 TWh but a median of only 12.46 TWh, confirming heavy right-skew driven by a handful of very large economies and global/regional aggregate rows.

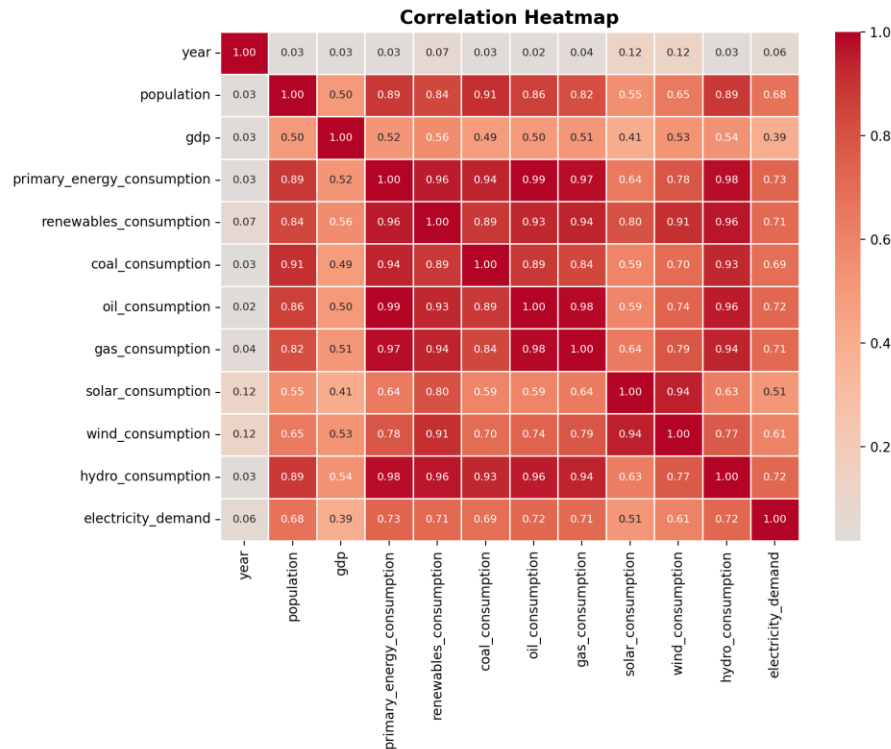


Figure 2: Correlation heatmap of numeric features. Most energy-consumption features and GDP show very strong positive correlation with electricity demand.

The correlation analysis showed that primary_energy_consumption ($r = 0.996$), gdp ($r = 0.994$), oil_consumption ($r = 0.979$), hydro_consumption ($r = 0.974$), renewables_consumption ($r = 0.971$), gas_consumption ($r = 0.968$), coal_consumption ($r = 0.933$), and population ($r = 0.899$) are all very strongly correlated with electricity demand. wind_consumption ($r = 0.819$) and solar_consumption ($r = 0.690$) show strong but comparatively weaker correlation, while year ($r = 0.064$) shows almost no linear relationship with demand on its own.

Key insights: Electricity demand scales almost linearly with overall primary energy consumption and GDP, which is expected since larger, wealthier economies consume more electricity. The very strong correlations among the predictors themselves (e.g., between primary_energy_consumption and the individual fuel-consumption columns) indicate significant multicollinearity, which regularized linear models such as Ridge Regression are well suited to handle. The extreme right-skew and outliers in the target (driven by rows for large countries and world aggregates) present a modelling challenge, since a small number of very large-demand records can dominate the error metrics.

2.3 Model Building

Task 1 - Neural Network

Architecture: A Multi-Layer Perceptron (MLPRegressor) with an input layer of 11 features, two hidden layers of 64 and 32 neurons using ReLU activation, and a single linear output neuron for the continuous target.

Training: Loss function: Mean Squared Error (MSE), implicitly optimized via the Adam solver with a learning rate of 0.001 and L2 regularization ($\alpha = 0.01$). The network was trained for up to 300 iterations with early stopping enabled and 10% of the training data reserved for internal validation, to reduce the risk of overfitting on this comparatively small feature set.

The Neural Network achieved a Training R^2 of 0.5805 and a Test R^2 of 0.4909, with a Training RMSE of 1,655.98 TWh and a Test RMSE of 1,900.90 TWh (Training MAE 481.77, Test MAE 509.75). The close agreement between training and test performance suggests the early-stopping regularization was effective at preventing overfitting, although the model's overall explanatory power remains moderate.

Task 2 - Classical ML Models

Model 1 - Ridge Regression: Linear regression with L2 regularization ($\alpha = 1.0$) applied to the standardized features, chosen to handle the multicollinearity identified during EDA while keeping the model simple and interpretable.

Model 2 - Random Forest Regressor: An ensemble of 300 decision trees ($\text{max_depth} = 25$, $\text{min_samples_split} = 5$, $\text{min_samples_leaf} = 2$, $\text{random_state} = 42$), chosen for its ability to capture non-linear relationships and interactions among the energy-consumption features without requiring feature scaling.

Model	Train MSE	Test MSE	Train RMSE	Test RMSE	Train MAE	Test MAE	Train R^2	Test R^2
Ridge Regression	2,874,762.89	3,626,770.49	1,695.51	1,904.41	537.39	559.64	0.560	0.489
Random Forest	1,427,305.76	2,557,374.99	1,194.70	1,599.18	213.41	305.73	0.782	0.640

Table 1: Baseline performance of the two classical regression models (all 11 features, default/manually chosen hyperparameters).

Even before tuning, the Random Forest baseline clearly outperformed Ridge Regression on every metric, indicating that the relationship between the predictors and electricity demand is not purely linear.

2.4 Model Evaluation

All models were evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). These are standard regression metrics: MAE and RMSE quantify average prediction error in the original units of electricity demand (TWh), while R^2 indicates the proportion of variance in electricity demand explained by each model. Reporting both training and test values for every metric allows overfitting to be diagnosed directly.

2.5 Hyperparameter Optimization

GridSearchCV with 5-fold cross-validation (scoring = R^2) was used to identify optimal hyperparameters for both classical models.

Ridge Regression: Tested alpha values [0.001, 0.01, 0.1, 1, 10, 100]. Best: alpha = 100, with a cross-validation R^2 of 0.5423.

Random Forest: Tested n_estimators [100, 200, 300], max_depth [15, 20, 25, None], min_samples_split [2, 5], and min_samples_leaf [1, 2]. Best: n_estimators = 100, max_depth = 15, min_samples_split = 5, min_samples_leaf = 1, with a cross-validation R^2 of 0.7138.

2.6 Feature Selection

SelectKBest with the f_regression scoring function was applied to rank the 11 available predictor features. Because the dataset contains only 11 predictors (fewer than the requested k = 15), all 11 features were retained; feature selection in this case therefore served to rank and interpret feature importance rather than to reduce dimensionality.

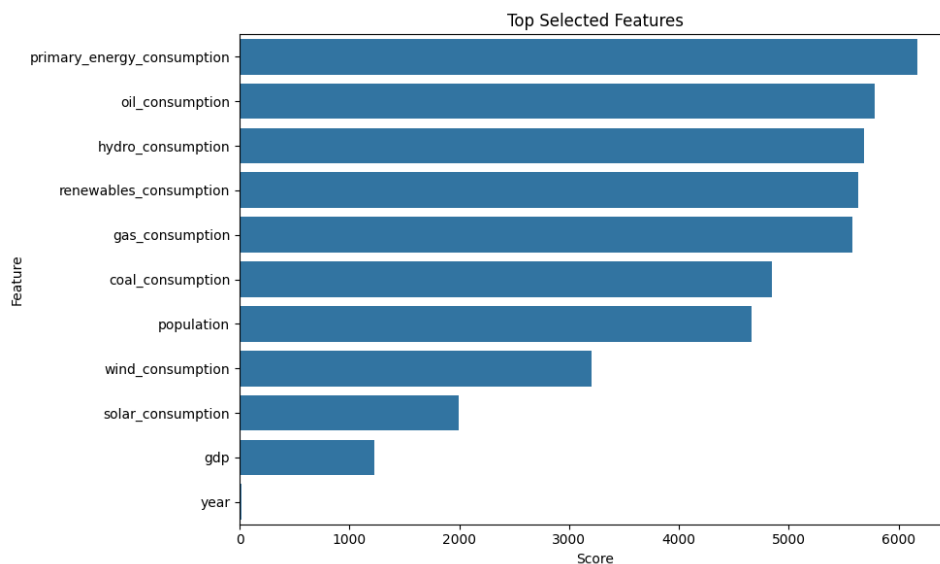


Figure 3: SelectKBest (f_regression) importance scores for all 11 predictor features.

Ranked by F-score: primary_energy_consumption (6,171.03), oil_consumption (5,786.23), hydro_consumption (5,681.45), renewables_consumption (5,634.76), gas_consumption (5,581.04), coal_consumption (4,846.22), population (4,657.41), wind_consumption (3,208.14), solar_consumption (1,992.04), gdp (1,228.78), and year (19.15). This ranking is consistent with the correlation analysis in Section 2.2 and confirms that year, on its own, contributes almost no explanatory power.

3. Results and Conclusion

3.1 Key Findings

Model	Features	Train MSE	Test MSE	Train RMSE	Test RMSE	Train MAE	Test MAE	Train R ²	Test R ²
Ridge (optimized)	All (11)	2,880,749	3,632,843	1,697.28	1,906.00	534.81	559.64	0.559	0.488
Random Forest (optimized)	All (11)	1,375,414	2,553,073	1,172.78	1,597.83	210.09	306.74	0.790	0.640

Table 2: Comparison of final regression models using optimal hyperparameters and the full (SelectKBest-ranked) feature set.

Both final models improved marginally over their respective baselines once optimal hyperparameters were applied. Random Forest remained the clear top performer, achieving a Test R² of 0.640 (explaining roughly 64% of the variance in electricity demand) versus 0.488 for Ridge Regression. The Neural Network (Test R² = 0.491, Section 2.3) performed similarly to Ridge Regression, suggesting that a simple linear model with regularization captures nearly as much signal as an unregularized MLP on this feature set, while the tree ensemble's ability to model non-linear interactions between energy-consumption features gives it a meaningful edge.

3.2 Final Model

The Random Forest Regressor (`n_estimators` = 100, `max_depth` = 15, `min_samples_split` = 5, `min_samples_leaf` = 1) is selected as the final model, achieving the highest test R² (0.640) and lowest test RMSE (1,597.83 TWh) and test MAE (306.74 TWh) among all three approaches. However, the gap between its training R² (0.790) and test R² (0.640) indicates some residual overfitting that a production deployment would need to address further.

3.3 Challenges

- Severe target missingness: 72.7% of rows lacked an `electricity_demand` value and had to be dropped, substantially reducing the usable sample from 23,377 to 6,378 records.
- Extreme target skew and outliers: electricity demand ranges from 0 to over 31,000 TWh, with world/regional aggregate rows creating extreme outliers that inflate error metrics such as RMSE.
- Multicollinearity: several energy-consumption features are very highly correlated with each other and with the target, complicating the interpretation of individual feature effects in the linear models.
- Feature selection did not reduce dimensionality, since the engineered feature set (11 columns) was already smaller than the intended `k` = 15, limiting the practical benefit of this step for this dataset.

3.4 Future Work

- Apply a log-transformation to the target variable to reduce the influence of extreme outliers and improve linear-model fit.

- Separate or flag world/regional aggregate rows from individual country rows, since mixing scales (a single country vs. "World") likely distorts model learning.
- Engineer additional features such as energy intensity (consumption per unit GDP) or per-capita consumption to better normalize across very differently sized economies.
- Explore gradient boosting methods (XGBoost, LightGBM) and additional regularization or pruning of the Random Forest to close the train-test performance gap.
- Expand hyperparameter search ranges and consider RandomizedSearchCV or Bayesian optimization for a larger search space within reasonable compute time.

4. Discussion

4.1 Model Performance

All three models achieved positive test R^2 scores in the 0.49-0.64 range, a substantial improvement over the earlier version of this project (which used only 63 records and produced negative R^2 for every model). This confirms that dataset size and target-variable quality were the dominant limiting factors previously, and that a larger, more representative sample (6,378 records) allows even simple models to capture meaningful structure. The Random Forest's superior performance (Test $R^2 = 0.640$) indicates that the relationship between macro energy/economic indicators and electricity demand includes non-linear effects and feature interactions that linear models (Ridge) and a shallow, lightly regularized MLP do not fully capture.

4.2 Impact of Hyperparameter Tuning and Feature Selection

Hyperparameter optimization produced a modest improvement in Random Forest's cross-validation R^2 relative to the manually chosen baseline configuration, and slightly reduced Ridge Regression's test error at $\alpha = 100$ compared with the unregularized baseline ($\alpha = 1$). Because all 11 engineered features were already highly informative (see Section 2.6), feature selection via SelectKBest did not remove any features in this case; its practical value here was limited to confirming that population and GDP-based features and all major fuel-consumption categories all carry genuine predictive signal, and that only the year column contributes negligible information on its own.

4.3 Interpretation of Results

Primary energy consumption, GDP, and oil consumption emerged as the strongest predictors of electricity demand, which is intuitive: larger and wealthier economies both consume more total energy and demand more electricity. Renewable, hydro, and gas consumption were similarly strong predictors, reflecting the fact that electricity-heavy economies tend to have diversified, large-scale energy portfolios rather than a specific correlation with clean energy per se. The weak standalone contribution of the year feature suggests that any long-term global trend in electricity demand is already captured indirectly through population and GDP growth, rather than through calendar time itself.

4.4 Limitations

The dataset mixes individual-country and aggregated (regional/world) observations within the same feature columns, which can distort the scale relationships that regression models learn. The extreme skew and heavy-tailed distribution of the target variable likely inflate RMSE for large-demand records, even though R^2 remains reasonable. The models are trained on a global cross-country panel without explicit country identifiers, meaning they cannot account for country-specific structural factors (climate, industrial mix, electrification rate) that influence demand beyond the included features. Finally, imputing missing predictor values with the median may understate the true variability of population, GDP, and energy consumption for the affected rows.

4.5 Suggestions for Future Research

Future research could incorporate country-level fixed effects or per-capita/intensity-normalized features to make the cross-country comparison more meaningful, apply target transformations (e.g., log or Box-Cox) to reduce the influence of outliers, and evaluate more powerful non-linear models such as gradient-boosted trees. Time-series-aware validation (e.g., grouping by country and validating on more recent years) would also give a more realistic estimate of how well such a model could forecast future electricity demand, rather than only interpolating within the historical range.

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